Stat. Analysis II

Chapter 7

Index Number

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Introduction

Business planning relies on analyzing operational trends. However, because individual product prices change at different rates, tracking the general price level is difficult. To solve this, **"Index Numbers"** are used as a statistical tool to visualize overall trends clearly.

Index Numbers

Index Numbers are statistical tools used to measure and compare changes or movements in data (such as prices, quantities, or values) over different time periods or locations. These changes are typically expressed as **ratios or percentages** relative to a designated **base period**.

Key Applications:

- **Business & Economics:** They serve as essential indicators of economic conditions, supporting decision-making, operational planning, and strategy.
- **Other Sectors (e.g., Education):** They function as development indicators to track progress and performance within a specific field.

Construction of Index Numbers

Key considerations for constructing an effective and accurate index:

1. **Purpose of Index Number Construction**: to establish clarity in analysis, which will assist in selecting appropriate items that meet requirements, comprehensively cover the subject matter, and ensure accuracy and reliability. Items may be divided into groups, categories, or according to similar characteristics.
2. **Data sources used in the analysis**: must be reliable and appropriately sized according to the type of index number. Additionally, the data collection process must be efficient.

3. **Reference Comparison or Base Year Determination**: The base year used for comparison must be appropriate:
   - It must be a period of normal economic conditions (free from wars, severe natural disasters, or policy crises).
   - It should not be too distant from the current period under study.

4. **Selection of Weights:** Weights must reflect reality since different items hold varying degrees of importance. For example:
   - For a **Price Index**, use **quantity** as the weight.
   - For a **Quantity Index**, use **price** as the weight.

   Note: Weights must be periodically updated to stay relevant to changing consumer behaviors over time.

Types of Index Numbers

Index numbers are widely used across various fields to track and examine changes. For instance, **educational indices** measure student continuation or dropout rates, while **population indices** track birth, death, and migration rates.

Classification by Purpose

1) **Price Index** is a value used to measure the relative change in prices of goods and services. Price indices are important in economics and business, such as the Wholesale Price Index and the Consumer Price Index.
2) **Quantity Index** is a value used to measure the relative change in quantities of goods and services. Important quantity indices include, for example, the Export Quantity Index.

3. **Value Index** is a value used to measure the relative change in the value of goods and services. The value of any commodity is obtained by multiplying the price per unit by the quantity. For example, if the export price of rice is 12,000 baht per ton and 500 tons of rice are exported, then the export value of rice = 12,000 × 500 = 6,000,000 baht. It can be observed that the value index reflects changes in both price and quantity simultaneously.
4. **Special-Purpose Index Numbers**

When classified by construction method, index numbers can be divided into two major categories:

1) Unweighted Index

2) Weighted Index

Notation Used in Index Number Calculation

Index number calculations employ various symbols as follows:

p = Price of goods

q = Quantity of goods

v = Value of goods

p_0 = Price of goods in the base period for comparison. For example, p_{2550} represents the price of goods in 2550, where 2550 is the base year.

q_0 = Quantity of goods in the base period for comparison. For example, q_{2550} represents the quantity of goods in 2550, where 2550 is the base year.

v_0 = Value of goods in the base period for comparison, where $v_0 = p_0 q_0$.

p_n = Price of goods in the period of interest for which the index number is calculated.

q_n = Quantity of goods in the period of interest for which the index number is calculated.

v_n = Value of goods in the period of interest for which the index number is calculated, where $v_n = p_n q_n$.

I_n = Price index in the period of interest for which the index number is calculated. For example, I_{2552} represents the price index in 2552.

QI_n = Quantity index in the period of interest for which the index number is calculated. For example, QI_{2552} represents the quantity index in 2552.

VI_n = Value index in the period of interest for which the index number is calculated. For example, VI_{2552} represents the value index in 2552.

Considerations in Constructing Index Numbers

Defining the Purpose of Construction

Since there are **various types of index numbers** and the **purpose of constructing** each type differs, it is essential to **clearly define the intended purpose** before construction. For example, the construction of a **consumer price index** aims to **measure changes in prices of goods and services necessary for living**.

Selecting Representative Items of Goods and Services for Index Number Calculation

The accuracy and reliability of an index number depend heavily on the selection of goods and services used as representatives in the calculation. They must adequately represent the entire market.

- **Categorization:** Representative items are organized into specific categories. For example, the Consumer Price Index (CPI) divides goods into 7 main categories (such as food and beverages, housing, and medical care).
- **Expert Selection:** Choosing specific items within each category requires expert knowledge to ensure the selected prices best reflect economic reality.

Selecting the Base Period for Comparison

The selection of a **base period** may be a year, month, or week. However, the base period **should be a normal period**, such as **one without war** conditions or with **overall normal economic conditions**. When comparing the period of interest to observe changes, the **time intervals should be equal**.

Selecting Formulas and Weighting Methods

Formula & Weighting Selection: The choice of formula and weighting method depends on the **constructor's judgment** to best fit the context, as formulas and weights are closely interrelated.

Classification by Number of Items:

- **Simple Index:** Calculated from a **single item** (e.g., a simple price index for just rice).
- **Composite Index:** Calculated from **multiple items** grouped together.

Classification by Method: Every index can be computed using either an **Unweighted** or a **Weighted** method.

Price index

The calculation of a price index is based on the principle of comparing the price of goods in the year of interest with the price of goods in the base year. The goods for which the price index is calculated may be a single item or multiple items combined.

An unweighted price index is calculated by giving equal importance to all items. Unweighted price indices are divided into three types: simple price index, simple aggregative price index, and simple average of relative price index, as follows:

1. Simple price index

A simple aggregative price index is a value that describes the change in prices of a group of goods in the year of interest compared to the prices of that group of goods in the base year. The calculation formula is as follows:

Simple Price Index $$(I_n) = \frac{p_n}{p_0} \times 100$$

Example 7.1 Suppose the price of a computer from a certain company in 2550 was 38,600 baht, and in 2551 the price was 32,000 baht. Find the simple price index for 2551 compared to 2550.

Solution:

1) The base year is 2550.

2) Calculate the simple price index as follows: $(I_n) = \dfrac{p_n}{p_0} \times 100$

$$I_{2551} = \frac{p_{2551}}{p_{2550}} \times 100$$

$$= \frac{32,000}{38,600} \times 100$$

$$= 82.9$$

3) This indicates that the computer price in 2551 decreased from the 2550 price by 17.1%.

2. Simple aggregative price index

A simple aggregative price index is a value that describes the change in prices of a group of goods in the year of interest compared to the prices of that group of goods in the base year. The calculation formula is as follows:

Simple Aggregative Price Index $$(I_n) = \frac{\sum p_n}{\sum p_0} \times 100$$

Example 7.2 Suppose that the prices of five consumer goods in the years 2551-2552 are shown in the table below. Find the simple aggregative price index for the year 2552 relative to 2551.

Product list	**Product price (Baht/Unit)**	
	2551	**2552**
Rice (kilogram)	15	18
Pork (kilogram)	90	95
Eggs size 0 (dozen)	35	40
Cooking oil (liter)	56	62
Sugar (kilogram)	20	22
Total	216	237

Solution:

1) The base year is 2551.

2) Calculate the simple aggregative price index as follows: $I_n = \frac{\sum p_n}{\sum p_0} \times 100$

$$I_{2552} = \frac{\sum p_{2552}}{\sum p_{2551}} \times 100$$

$$= \frac{237}{216} \times 100$$

$$= 109.72$$

3) This result indicates that the overall prices of the five consumer goods in 2552

increased from 2551 by 9.72%.

3. Simple average of relative price index

The Simple Average of Relative Price Index works by calculating the relative price change of each item first, then summing them up to find the average. This specific method effectively eliminates the problem of different price magnitudes, preventing expensive items from distorting the results. Consequently, it reflects true relative price changes, making it far more accurate and reliable than the Simple Aggregative Price Index.The calculation formula is as follows:

Simple Average of Relative Price Index $$(I_n) = \frac{\sum \frac{p_n}{p_0}}{n} \times 100$$

Example 7.3 Using the data from Example 7.2, find the simple average of the relative price index.

Solution:

1) The base year is 2551.

2) Calculate the simple average of the relative price index as follows: $I_n = \dfrac{\sum \frac{p_n}{p_0}}{n} \times 100$

therefore $I_{2552} = \dfrac{\sum \frac{p_{2552}}{p_{2551}}}{5} \times 100$

3) Calculate the various values as follows:

Product list	Product price (Baht/Unit)		$\frac{p_{2552}}{p_{2551}}$
	2551	**2552**	
Rice (kilogram)	15	18	$\frac{18}{15}=1.20$
Pork (kilogram)	90	95	$\frac{95}{90}=1.06$
Eggs size 0 (dozen)	35	40	$\frac{40}{35}=1.14$
Cooking oil (liter)	56	62	$\frac{62}{56}=1.11$
Sugar (kilogram)	20	22	$\frac{22}{20}=1.10$
Total	216	237	5.61

4) Calculate the value $I_{2552} = \frac{5.61}{5} \times 100 = 112.11$

5) This indicates that the prices of these 5 consumer goods in 2552 increased from 2551 by 12.11%.

Quantity index

A quantity index is a value that indicates the change in quantities of goods and services in the year of interest compared to the base year. The quantity index is used to measure this change and is particularly useful in industry, as changes in product quantities affect production planning. The construction of quantity indices follows principles similar to price indices, except that price is replaced by quantity.

An unweighted quantity index follows the same principle as an unweighted price index, which is calculating the index by giving equal importance to all items. Unweighted quantity indices are divided into three types:

1. Simple quantity index

A simple quantity index is a value that describes the change in the quantity of a single item in the year of interest compared to the quantity in the base year. The calculation formula is as follows:

$$\textbf{Simple Quantity Index} \quad (QI_n) = \frac{q_n}{q_0} \times 100$$

2. Simple aggregative quantity index

A simple aggregative quantity index is a value that describes the change in quantities of a group of goods in the year of interest compared to the quantities of that group of goods in the base year. The calculation formula is as follows:

$$\textbf{Simple Aggregative Quantity Index} \quad (QI_n) = \frac{\sum q_n}{\sum q_0} \times 100$$

3. Simple average of relative quantity index

A simple average of relative quantity index is a value that describes the change in quantities of a group of goods in the year of interest compared to the quantities of that group of goods in the base year. The relative quantity of each individual item is calculated first, then the relative quantities of all items are summed together and averaged by dividing by the number of items. The calculation formula is as follows:

Simple Average of Relative Quantity Index $$(QI_n) = \frac{\sum \frac{q_n}{q_0}}{n} \times 100$$

Example 7.4 The data in the table below shows the employment numbers for each of 4 sectors over a 2-year period, from 2550 to 2551, as follows:

Sector	**Employment Quantity (persons)**	
	2550	**2551**
Agriculture	16,031	15,945
Manufacturing Industry	4,357	4,178
Mining	44	40
Construction	2,021	1,763
Total	22,453	21,926

a) Calculate the simple aggregative quantity index for employment in 2551 using 2550 as the base year.

b) Calculate the simple average of relative quantity index for employment in 2551 using 2550 as the base year.

Solution:

a) Since 2550 is used as the base year, the simple aggregative quantity index is calculated as follows:

$$QI_n = \frac{\sum q_n}{\sum q_0} \times 100$$

$$QI_{2551} = \frac{\sum q_{2551}}{\sum q_{2550}} \times 100$$

$$= \frac{21,926}{22,453} \times 100$$

$$= 97.65$$

This indicates that total employment in 2551 decreased from 2550 by 2.35%.

b) Since 2550 is used as the base year, the simple average of relative quantity index is calculated as follows:

$$QI_n = \frac{\sum \frac{q_n}{q_0}}{n} \times 100$$

$$QI_{2551} = \frac{\sum \frac{q_{2551}}{q_{2550}}}{4} \times 100$$

Calculate the various values as follows:

Sector	Employment Quantity (persons)		$\frac{q_{2551}}{q_{2550}}$
	2550	2551	
Agriculture	16,031	15,945	$\frac{15,945}{16,031} = 0.995$
Manufacturing Industry	4,357	4,178	$\frac{4,178}{4,357} = 0.959$
Mining	44	40	$\frac{40}{44} = 0.909$
Construction	2,021	1,763	$\frac{1,763}{2,021} = 0.872$
Total	22,453	21,926	3.735

$$QI_{2551} = \frac{3.735}{4} \times 100 = 93.38$$

This indicates that total employment in 2551 decreased from 2550 by 6.62%.

Value index

A value index is a value that measures the change in the value of goods and services from one period to another, such as the change in sales value of a certain product from 2551 to 2552. The calculation of a value index uses the same principle as that used in calculating a price index.

From the value of goods $v = pq$ and the total value of goods $\sum v = \sum pq$, the simple aggregative value index is the relative ratio that shows the change in total value of goods in the year of interest compared to the base year. The calculation formula is summarized as follows:

Simple Aggregative Value Index $(VI_n) = \dfrac{\sum p_n q_n}{\sum p_0 q_0} \times 100$

In cases where it is desired to calculate a weighted aggregate value index, it is called a weighted average of relative value index, with the following calculation formula:

Weighted Average of Relative Value Index

$$(VI_n) = \frac{\left(\sum \frac{p_n q_n}{p_0 q_0} \times v\right)}{\sum v} \times 100$$

by $v = \begin{cases} p_0 q_0 & \text{when using base year value as weight} \\ p_n q_n & \text{when using current year value as weight} \\ p_a q_a & \text{when using any year's value as weight} \end{cases}$

Example 7.5 The table below shows data on pharmaceutical sales for disease prevention and treatment for 2549 and 2551. Calculate the weighted average of the relative value index when the base year value of 2549 is designated as the weight.

Medicine	Price (Baht)		Quantity (tablets)	
	2549	2551	2549	2551
Penicillin	5	9	25,000	29,500
Streptomycin	10	12	18,000	24,000
Alkaloid	8	13	13,400	17,000

Solution:

1) Since the base year value of 2549 is designated as the weight,

$$v = v_0 = p_0 q_0 = p_{2549} q_{2549}$$

2) Calculate the weighted average of relative value index as follows:

$$VI_n = \frac{\left(\frac{\sum p_n q_n}{\sum p_0 q_0} \times v\right)}{\sum v} \times 100$$

$$VI_{2551} = \frac{\left(\sum \frac{p_{2551} q_{2551}}{p_{2549} q_{2549}} \times p_{2549} q_{2549}\right)}{\sum p_{2549} q_{2549}} \times 100$$

3) Calculate the various values as follows:

Medicine	$p_{2551}q_{2551}$	$p_{2549}q_{2549}$	$\sum \frac{p_{2551}q_{2551}}{p_{2549}q_{2549}} \times p_{2549}q_{2549}$
Penicillin	265,500	125,000	$\frac{265,500}{125,000} \times 125,000 = 265,500$
Streptomycin	288,000	180,000	$\frac{288,000}{180,000} \times 180,000 = 288,000$
Alkaloid	221,000	107,200	$\frac{221,000}{107,200} \times 107,200 = 221,000$
Total		412,200	774,500

4) Therefore, $VI_{2551} = \frac{774,500}{412,200} \times 100 = 187.89$

5) This indicates that the sales value of pharmaceuticals in 2551 increased from 2549 by 87.89%.

Consumer Price Index: CPI

The consumer price index, or cost of living index, is a tool used to measure changes in the prices of goods and services that most consumers regularly purchase in fixed quantities and quality as defined in the base year. Consumers refer to households with regular income or salaries, such as employees and civil servants.

The goods and services used to calculate the consumer price index are divided into 7 categories as follows:

1) Food and Beverages: this includes rice, flour, flour products, meat, duck, chicken, aquatic animals, vegetables, fruits, eggs, milk, dairy products, food purchased from markets, and non-alcoholic beverages.

2) Housing: this includes accommodation costs, house rent, construction costs, household furnishings, electricity, water supply, fuel, household textiles, and household cleaning materials.

3) Clothing and Footwear: this includes clothing for men and boys, clothing for women and girls, and tailoring services.

4) Medical Care and Personal Services: this includes medical examination fees, medicine costs, and personal service fees.

5) Transportation and Communication

6) Recreation, Reading, and Education

7) Tobacco and Alcoholic Beverages

- **Producer & Methodology:** The Ministry of Commerce calculates the CPI by surveying consumption quantities from representative households in the base year. Retail prices for the 7 categories of goods and services are collected from markets across Bangkok, its vicinity, and all 4 regions of Thailand.
- **Key Benefits:** Serves as a general economic indicator and measures consumer purchasing power (used to calculate the purchasing power of money).
- **Applications:** Used as a guideline for both government and private sectors to adjust salaries, wages, minimum wages, rent, and social assistance benefits.

Benefits of the Consumer Price Index (CPI)

The CPI is used as a tool to measure the cost of living and monetary conditions, specifically the **inflation rate** (for instance, a CPI of 154 in 1995 compared to a base year of 1986 indicates a total inflation of 54%, averaging about 6% per year). Additionally, it measures the **purchasing power of money**—reflecting its real value under current economic conditions—and is applied to calculate **real income**, using several key formulas.

$$Purchasing\ Power\ of\ Money = \frac{1}{Consumer\ Price\ Index} \times 100$$

$$Real\ Income = \frac{Current\ Monetary\ Income}{Consumer\ Price\ Index} \times 100$$

Furthermore, the consumer price index is used as a deflator (statistical deflation). Since value is the product of price and quantity, value may change due to price alone or due to quantity alone. Therefore, values can be adjusted according to the desired price level as follows:

$$Value\ at\ Desired\ Price\ Level = \frac{Current\ price\ value}{Consumer\ Price\ Index} \times 100$$

Example 7.6 If the consumer price index and monthly income of Mr. Tawan from 2548-2552 are as shown in this table:

Year	2548	2549	2550	2551	2552
Income (Baht)	10,200	10,700	11,600	12,800	13,700
Consumer Price Index	100	105	110.2	114.8	120.5

a) Calculate the purchasing power for each year when 2548 is designated as the base year.

b) Find Mr. Tawan's real income for each year when 2548 is designated as the base year.

c) What should Mr. Tawan's monthly income be in 2552 to maintain the same purchasing power as in 2548?

d) If Mr. Tawan purchased a house in 2548 for 900,000 baht and wants to sell the house in 2552, at what price should he sell it to maintain the same value?

Solution:

a) From the formula:

$$Purchasing\ Power\ of\ Money = \frac{1}{Consumer\ Price\ Index}$$

Calculate the purchasing power for each year as follows:

Year	Consumer Price Index (Base Year 2548)	Purchasing Power of Money
2548	100	$\frac{1}{100} \times 100 = 1$
2549	105	$\frac{1}{105} \times 100 = 0.95$
2550	110.2	$\frac{1}{110.2} \times 100 = 0.91$
2551	114.8	$\frac{1}{114.8} \times 100 = 0.87$
2552	120.5	$\frac{1}{120.5} \times 100 = 0.83$

The calculated purchasing power can be interpreted as follows: In 2548, the purchasing power equals 1, meaning 1 baht has a purchasing power equal to 1 baht. In 2549, the purchasing power equals 0.95, meaning 1 baht has the purchasing power equivalent to 0.95 baht when compared to 2548, and so on.

b) From the formula:

$$Real\ Income = \frac{Current\ Monetary\ Income}{Consumer\ Price\ Index} \times 100$$

Calculate the real income as follows:

Year	Consumer Price Index (Base Year 2548)	Nominal Income (Baht)	Real Income (Baht)
2548	100	10,200	$\frac{10,200}{100} \times 100 = 10,200$
2549	105	10,700	$\frac{10,700}{105} \times 100 = 10,190.48$
2550	110.2	11,600	$\frac{11,600}{110.2} \times 100 = 10,526.32$
2551	114.8	12,800	$\frac{12,800}{114.8} \times 100 = 11,149.83$
2552	120.5	13,700	$\frac{13,700}{120.5} \times 100 = 11,369.29$

The calculated real income can be interpreted as follows: In 2548, the real income equaled 10,200 baht, which equals the nominal income because 2548 is used as the base year. In 2549, the real income equaled 10,190.48 baht. It can be observed that although the nominal income in 2549 increased from 2548 by 10,700 - 10,200 = 500 baht, the rate of income increase was less than the rate of increase in consumer goods prices. This resulted in real income actually decreasing by 10,200 - 10,190.48 = 9.52 baht. The real income for other years can be explained in a similar manner.

c) Let S represent the monthly income in 2552.

Purchasing Power of Money in 2548 = Purchasing Power of Money in 2552

$$\frac{1}{100}\times 100\times 10,200=\frac{1}{120.5}\times 100\times S$$

$$S=\frac{120.5}{100}\times\frac{100}{100}\times 10,200$$

$$=\frac{120.5}{100}\times 10,200$$

$=12,291$ baht

Therefore, Mr. Tawan should have a monthly income of 12,291 baht in order to maintain the same purchasing power as in 2548.

d) The consumer price index for 2552 compared to 2548 equals 120.5. From the formula:

$$Value\ at\ Desired\ Price\ Level = \frac{Current\ price\ value}{Consumer\ Price\ Index} \times 100$$

$$900{,}000 = \frac{Current\ price\ value}{120.5} \times 100$$

Therefore, the house should be sold at $= \dfrac{900{,}000 \times 120.5}{100} = 1{,}084{,}500$ baht